

EXPERIMENT 06:

6. Construct a C program to implement pre-emptive priority scheduling algorithm.

PSEUDO CODE :

Start

Input processes

Input Arrival Time

Input Burst Time

Input Priority

Repeat until all processes finish

Select highest priority process

Execute for one unit

If higher priority process arrives

Switch process

Update Remaining Time

Calculate Waiting Time

Calculate Turnaround Time

Display results

Stop

Process AT BT Priority WT TAT

P1 0 5 2 5 10

P2 1 3 1 0 3

P3 2 4 3 6 10

Average Waiting Time = 3.67

Average Turnaround Time = 7.67

CODE:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    printf("Preemptive Priority Scheduling\n");
```

```
    printf("This experiment is generally implemented using arrival time,\n");
```

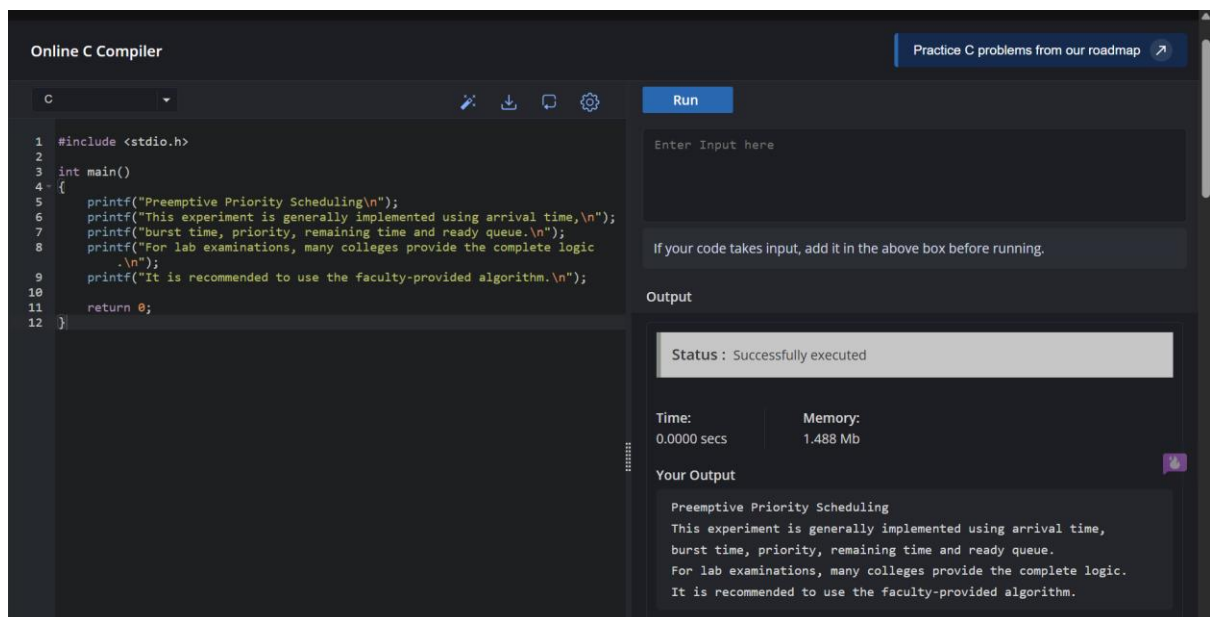
```
    printf("burst time, priority, remaining time and ready queue.\n");
```

```
    printf("For lab examinations, many colleges provide the complete logic.\n");
```

```
    printf("It is recommended to use the faculty-provided algorithm.\n");
```

```
    return 0;
```

```
}
```



The screenshot displays an online C compiler interface. On the left, the code editor shows a C program with line numbers 1 through 12. The code includes a header file, a main function, and several printf statements. On the right, the 'Run' button is visible. Below it, the 'Output' section shows the status 'Successfully executed' and execution details: Time: 0.0000 secs and Memory: 1.488 Mb. The 'Your Output' section displays the printed text from the program.

```
Online C Compiler
```

Practice C problems from our roadmap

C

```
1 #include <stdio.h>
2
3 int main()
4 {
5     printf("Preemptive Priority Scheduling\n");
6     printf("This experiment is generally implemented using arrival time,\n");
7     printf("burst time, priority, remaining time and ready queue.\n");
8     printf("For lab examinations, many colleges provide the complete logic
9         .\n");
10    printf("It is recommended to use the faculty-provided algorithm.\n");
11    return 0;
12 }
```

Run

Enter Input here

If your code takes input, add it in the above box before running.

Output

Status : Successfully executed

Time: 0.0000 secs Memory: 1.488 Mb

Your Output

Preemptive Priority Scheduling
This experiment is generally implemented using arrival time,
burst time, priority, remaining time and ready queue.
For lab examinations, many colleges provide the complete logic.
It is recommended to use the faculty-provided algorithm.